FABRID-IDS: Federated Alert-Budget Reallocation for Heterogeneous IoT Intrusion Detection

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Abstract

The same anomaly scores can yield strong pooled ranking metrics but unreliable alerts after thresholds are selected for individual IoT devices. FABRID-IDS examines the allocation of a federation-wide nominal false-positive-rate (FPR) budget after training. Client-level targets are chosen jointly, then converted into thresholds using separate benign calibration records. The main evaluation covers nine N-BaIoT devices, ten detector seeds and five budget levels. A secondary experiment uses 19 CIC IoT-DIAD clients with a different feature representation and scorer. At $\boldsymbol{B = 0.001}$, both unrestricted allocation objectives (Macro and Tail) produce a held-out federation FPR of 0.00170, approximately 1.696 times the budget. Conservative calibration lowers that FPR to 0.00036, but macro recall decreases from 0.3778 to 0.2445. Equal-FPR performs better in macro recall at several nominal budgets, although its measured FPR also exceeds the specified budget at several operating points. Pooled AUROC and AUPRC reach 0.9995 and 0.9999; nevertheless, the minimum client recall remains zero for every evaluated FABRID-family policy and budget on N-BaIoT. These results expose a practical limitation: allocating nominal FPR targets is not sufficient to control measured federation false alarms or maintain detection for every device.

Keywords: federated intrusion detection, Internet of Things, alert-budget reallocation, false-positive-rate constraint, threshold calibration

1 Introduction

In a federated IoT intrusion-detection system, training produces an anomaly score, but an alert is issued only after that score crosses a threshold. The choice of threshold becomes difficult when participating devices have different benign-score distributions or when their attacks have different score profiles. Existing work examines federated detector training and both shared and device-specific thresholds [1–6]. It leaves a separate operational question: if the federation permits a limited average false-positive rate, which devices should receive larger or smaller nominal allowances?

FABRID-IDS addresses that question after the scoring model is fixed. Using labeled validation attacks and benign frontier scores, the allocator chooses nominal client rates α_k subject to $\sum_k w_k \alpha_k \leq B$. Each device subsequently determines its alert threshold from a different benign calibration partition. The held-out evaluation therefore tests both the allocation decision and what happens when those rates are implemented using finite local calibration samples. The shared budget B restricts the nominal assignment; it is not a direct constraint on false alarms observed later.

Three aspects are evaluated: the allocation of different nominal FPR allowances under one federation constraint; the gap between allocated rates and the thresholds produced by independent calibration; and the effect of each policy on average and worst-client attack recall. The N-BaIoT policies are compared on the same frozen scores and detector seeds.

2 Related work and scope

The distinction between detector training and alert policy is important when positioning FABRID-IDS. Federated IoT IDS papers have primarily developed learning architectures and aggregation strategies [1–3, 7]. Other work considers anomaly thresholds and detection at distributed edges [8, 9]. Laridi et al. [4] combine client-side summary statistics to calculate a shared anomaly threshold. Asiri et al. [5] derive local benign reconstruction-error quantiles and average the resulting thresholds. Fed-DTCN [6] instead uses benign hold-out samples to determine client-specific thresholds at prescribed local FPRs. These methods provide relevant thresholding baselines or context, but do not make the federation-wide division of nominal FPR allowances their optimization variable.

Error-rate constraints also appear in Neyman–Pearson classification [10], while conformal methods use calibration ranks to support finite-sample statements under explicit sampling assumptions [11]. Federated conformal work studies a related but different problem involving predictor sharing and calibration-score aggregation [12]. FABRID-IDS takes the calibration constructions as existing tools and places the allocation decision before them: the optimizer chooses the nominal target for each client, and a separate local sample determines the final threshold. The contribution is this federation-constrained allocation protocol and its measured behavior, not a new order-statistic theorem or a guarantee of realized FPR.

Training-time fairness methods and agnostic federated learning modify the learning objective or the influence of client data [7]. FABRID-IDS does neither. Once training is complete, adjusting α_k moves the decision threshold for device k without changing

its benign or attack score rankings. This matters experimentally: threshold policies can be evaluated using identical stored scores, while any weaknesses of the scorer remain common to them. An allocation scheme cannot recover an attack that the fixed scorer consistently ranks below the required threshold without allowing a different operating point.

Three operating choices are therefore kept distinct. A global absolute threshold applies one score cutoff everywhere. Equal-FPR allows local thresholds but specifies the same nominal rate at every client. FABRID-IDS also calibrates locally, while allowing the assigned rates to differ under the weighted federation constraint.

3 FABRID-IDS

3.1 Protocol and utility

Let $s_k(\cdot)$ be the frozen anomaly scorer for client k. Benign records are divided according to their released file order: 50% for training, 20% for estimating the allocation frontier, 10% for final threshold calibration and 20% for testing. For every retained attack subtype, the initial 20% is reserved for validation and the rest for testing. Only the frontier benign records and attack-validation subset are available when the allocator chooses nominal rates. The final-calibration subset is used afterward to turn those rates into thresholds, and the test subset is left untouched until evaluation. These proportions describe the available benchmark protocol, not a confirmed chronological deployment split.

For a benign calibration sample of size n, sort its scores as $x_{(1)} \leq \cdots \leq x_{(n)}$. The target rate α is mapped to a threshold using the order-statistic rule in Eq. (1) [11].

$$\tau(\alpha) = x_{(r)}, \qquad r = \lceil (n+1)(1-\alpha) \rceil , \tag{1}$$

Set $\tau = +\infty$ for $\alpha \leq 0$, $n = 0$, or $r > n$; otherwise an observation is flagged only when its score is strictly greater than τ. The rule is applied first to the frontier benign sample at each candidate rate. The provisional threshold is then used to calculate $u_k(\alpha)$, the mean validation recall over the eligible attack subtypes of client k. Attack labels are therefore required to construct the utility curve. The search grid contains 207 rates between 0 and 0.05 and includes all five evaluation budgets. Subtypes need at least 50 validation records, and a client is eligible only with at least 200 such records across two subtypes.

Model fitting and feature normalization use the benign training split of each client. Once an allocation has been selected, its provisional frontier thresholds are discarded and the order-statistic procedure is repeated on final-calibration benign scores. The test scores are not used at either selection stage. This separation prevents direct reuse of frontier observations for final threshold setting, but it does not force the frontier, calibration and test partitions to have identical score distributions.

Nominal target rates are more finely spaced than the thresholds obtainable from a finite benign sample. Several adjacent values of α_k may select the same order statistic, yielding the same threshold even though the optimizer records different rates. In that case, held-out decisions, recall and FPR also coincide because the scorer and test data

are fixed. An allocated-rate change should consequently be interpreted as a distinct operating choice only if it produces a distinct calibrated cutoff.

Attack recall is first computed separately for eligible subtypes and averaged within each device. The federation macro-recall endpoint gives each eligible device equal weight, regardless of the number of retained attack rows. The worst-client endpoint is the minimum of these device-level macro recalls. Both are reported with the held-out benign FPR.

3.2 Allocation and calibration

The allocator selects one nominal rate from the candidate grid for each eligible client. Those choices are feasible only when the weighted sum is at most the specified federation budget:

$$\sum_{k=1}^{K} w_k \alpha_k \leq B. \tag{2}$$

The N-BaIoT implementation sets $w_k = 1/K$, so the budget represents an equal-device average rather than a row-weighted pooled rate. Allocation is solved as a one-hot mixed-integer program with HiGHS. The Macro objective maximizes the mean estimated client recall u_k; Tail instead maximizes the mean of the lowest $q = \lceil 0.25K \rceil$ client utilities [13]. With nine devices, $q = 3$. The Tail objective thus averages the three weakest utilities rather than taking only the single minimum, and it is not numerically identical to empirical CVaR with a 0.25 tail fraction, which places a fractional weight on the third observation [13]. Each frontier sample has finite FPR resolution. For n_f benign frontier records, positive candidate rates below $2/(n_f + 1)$ are removed because they cannot create a positive empirical calibration-alert rate with the strict threshold rule.

The equal-device definition should not be confused with pooling all benign records and calculating one false-positive fraction. Moreover, Eq. (2) defines a maximum expenditure, not a requirement to spend the full budget. The optimizer may leave part of B unused or set a client target to zero when the estimated utility does not justify spending additional allowance on that device.

Each optimization follows a specified order of priorities. Macro first maximizes estimated mean recall, then selects the smallest nominal expenditure among solutions accepted as equivalent under the solver tolerances. Tail places the lowest-three-client mean first, the overall estimated mean second and nominal expenditure third. Estimated utility variance and deterministic tie-breaking resolve the remaining choices. These calculations use frontier scores and labeled validation attacks only. Equal-FPR assigns B to every client, and all three unrestricted policies use the same final-calibration benign samples and Eq. (1). For the conservative Macro and Tail variants, the client allocations are retained but the final threshold ranks are determined by Eq. (3).

A higher Tail objective does not necessarily imply better protection for the single weakest client. For example, an increase in the estimated recall of two members of the three-device lower group can improve its mean even if the remaining member continues to miss every evaluated attack.

For conservative calibration, the following binomial criterion is used to choose the final threshold rank [10]:

$$r_{.95} = \min\{r : \Pr[\text{Binomial}(n, 1-\alpha) \leq r-1] \geq 0.95\}. \tag{3}$$

For each client, the conservative procedure searches integer ranks from 1 to n and uses the first rank meeting the binomial condition in Eq. (3). If no rank qualifies, or if $n = 0$ or $\alpha = 0$, the threshold is infinite and no alerts are issued. Alerts otherwise require $s > \tau$, as in ordinary calibration. Under independently and identically distributed (i.i.d.) benign calibration and future scores, the rank construction provides a one-sided 95% population-FPR confidence statement for an individual client [10]. That statement concerns repeated calibration sampling; it is not a guarantee about the empirical FPR of the fixed, source-ordered test partition used here. The conservative variant is therefore evaluated as a sensitivity check rather than presented as validated federation-level control.

Equation (3) generally requires a larger calibration rank than Eq. (1). A larger rank moves the threshold upward, making benign alerts less frequent but also allowing more attacks to fall below the cutoff. Neither method looks at test labels before choosing the threshold, and neither can compensate automatically if benign-score behavior differs between calibration and testing.

The allocation constraint and the reported test outcome answer different questions. A client can exceed its nominal rate on held-out data even when Eq. (2) was respected at allocation time. Equally, a low held-out FPR in one partition does not establish that the same threshold will remain safe on later traffic. For this reason, four quantities are kept separate: selected nominal client rate, final calibration rank, FPR within the calibration sample, and FPR on the held-out benign sample. Ordinary Macro, Tail and Equal-FPR differ in the assigned rates, not in the records or calibration formula used to turn those rates into operating thresholds.

The smallest nonzero calibration-sample FPR is $1/n$ when the n benign scores have no relevant ties. Because alerts use a strict greater-than comparison, targets below $2/(n+1)$ either produce an out-of-range rank or select the maximum observed benign score; both cases yield zero calibration alerts. Consequently, a grid of finely spaced target rates contains choices with identical zero-alert behavior. The final-calibration subset for Ecobee has 1,311 benign records, which places this transition at approximately 0.001524. A nominal rate below that value cannot yield a positive empirical FPR on its calibration subset, although the same threshold may still flag benign observations in the separate test subset.

3.3 Frozen scorer

Each N-BaIoT device uses a frozen hybrid anomaly score constructed from two components: autoencoder reconstruction error and the temporal jitter feature `HH_jit_L0.1_mean` (f_{75}). Before combining them, both components are expressed as percentile ranks against fixed frontier references; the final score is their maximum. The f_{75} feature and this scorer configuration were selected using the predeclared development-only ablations, not retuned for each threshold policy. The autoencoder

has hidden layers [64, 16] and is trained by federated averaging for ten rounds, with three local epochs per round. Optimization uses Adam at learning rate 0.001, batch size 64 and mean-squared-error loss. Aggregation is weighted by the available training-row counts, while each client's normalization parameters come from its own training partition. Ten detector seeds provide the paired repetitions. Every evaluated threshold policy receives exactly the same scores within a seed; hence changes in threshold can change recall and FPR but not score-ranking measures such as AUROC or AUPRC.

4 Experimental methodology

N-BaIoT is the main evaluation dataset. Its nine devices serve as the natural client units, and each is tested at five nominal federation budgets: $B = 0.001$, 0.0025, 0.005, 0.010 and 0.020 (Table 1). The retained traffic includes benign observations and attacks from the BASHLITE and Mirai families [14]. Device-level sample sizes differ substantially: the benign count is 13,113 for Ecobee and 175,240 for the Philips baby monitor. Since final calibration receives 10% of each device's benign rows, those unequal file sizes produce unequal calibration precision. Macro recall is averaged over eligible attack subtypes within each device and then across devices; the minimum-client endpoint records the lowest device-level value.

For any one detector seed, all policies are applied to identical trained models and score partitions. Comparing Macro against Equal-FPR, for example, therefore produces a within-seed difference rather than a comparison between unrelated detector runs. Ten seeds are available, and each contributes one observation for an endpoint at a given budget. The mean across seeds characterizes detector initialization and training variation on the fixed benchmark; it cannot be interpreted as a mean across newly collected clients, different collection periods or independently sampled traffic.

Held-out federation FPR uses the same device weights $w_k = 1/K$ as Eq. (2), applied to the benign test FPR measured separately for every device. This makes the nominal constraint and reported federation endpoint comparable in their weighting, although they are not guaranteed to have equal values. The test partition is derived from source-ordered files whose features were constructed from overlapping windows. It is not known to be an independent sample from the same distribution as calibration. Repeating detector training with different random seeds does not remove this dataset limitation.

Two comparisons were specified for N-BaIoT: Macro minus Equal-FPR on federation macro recall, and Tail minus Equal-FPR on minimum-client recall. At each budget, the endpoint is evaluated separately for every trained seed before the baseline value is subtracted. The resulting ten differences preserve the pairing between policies. Traffic rows and attack subtypes are inputs to the client-level endpoint; they do not become additional statistical replicates for the tests.

Inference uses a two-sided exact paired sign-flip test, enumerating all $2^{10} = 1{,}024$ sign assignments for the ten seed-level differences. A 95% pointwise confidence interval is also calculated using 50,000 bootstrap draws of those paired differences with replacement. The five budget-specific p-values within each endpoint family undergo Holm correction; the bootstrap intervals are not adjusted for simultaneous coverage. These quantities describe variation across the ten trained scorers on the same retained dataset. They

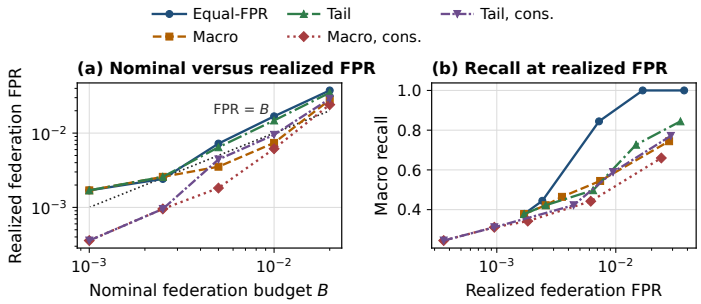


Fig. 1 N-BaIoT ten-seed means: realized FPR versus nominal budget (left) and macro recall versus realized FPR (right). The diagonal marks equal rates.

do not estimate uncertainty for new device populations, newly collected benign traffic or repeated calibration periods.

CIC IoT-DIAD, originating from the device-identification and anomaly-detection work of Rabbani et al. [15], is used as a secondary protocol check. Its 19 physical devices contribute 57 features, and a separate detector is trained for this dataset using reconstruction error rather than the N-BaIoT hybrid scorer. Benign partitions and the fraction of attacks reserved for validation follow the primary experiment. No N-BaIoT checkpoint or score ranking is transferred. Because the dataset, set of clients, representation and scorer all change together, this branch can show whether an FPR mismatch also occurs elsewhere, but it cannot determine which changed factor caused the difference.

5 Results

5.1 Nominal and realized operating points

At $B = 0.001$, unrestricted Macro and Tail both produce a realized federation FPR of 0.00170 ($1.696B$). Their conservative variants reduce it to 0.00036 ($0.358B$), while macro recall falls from 0.3778 to 0.2445. All four allocations satisfy the same nominal budget ceiling, but the realized FPR is measured independently on held-out benign records after local calibration.

The held-out FPR does not track the nominal budget in the same way for every policy. At $B = 0.0025$, Macro and Tail operate at approximately 1.03B, only slightly above the allocated ceiling. Equal-FPR also overshoots at several budgets, showing that the mismatch is not unique to optimized rate allocation. Figure 1(a) shows the deviation from the nominal diagonal, while Fig. 1(b) places each measured macro-recall value against its corresponding realized FPR.

At nominal budget $B = 0.005$, the measured macro recalls are 0.8444 for Equal-FPR, 0.4644 for Macro and 0.4977 for Tail. Their realized federation FPRs are 1.45B, 0.71B and 1.29B, respectively.

Allocation stability was checked by rerunning unrestricted Macro after bootstrap resampling the frontier benign scores and validation attack scores within each subtype. There are 500 such resamples per seed and budget. The trained scorer, the separate final-calibration scores and the test scores remain unchanged. For each seed–client–budget

Table 1 N-BaIoT ten-seed means. Each policy cell reports macro recall; realized federation FPR divided by nominal budget. All FABRID-family cells have zero minimum-client recall.

B	Equal-FPR	Macro	Tail	Macro, cons.	Tail, cons.
0.0010	0.3778; 1.70	0.3778; 1.70	0.3778; 1.70	0.2445; 0.36	0.2445; 0.36
0.0025	0.4445; 0.97	0.4222; 1.03	0.4222; 1.03	0.3111; 0.38	0.3111; 0.38
0.0050	0.8444; 1.45	0.4644; 0.71	0.4977; 1.29	0.3422; 0.36	0.4222; 0.88
0.0100	0.9999; 1.68	0.5444; 0.74	0.7266; 1.48	0.4422; 0.62	0.5889; 0.95
0.0200	1.0000; 1.88	0.7444; 1.40	0.8444; 1.75	0.6600; 1.21	0.7711; 1.45

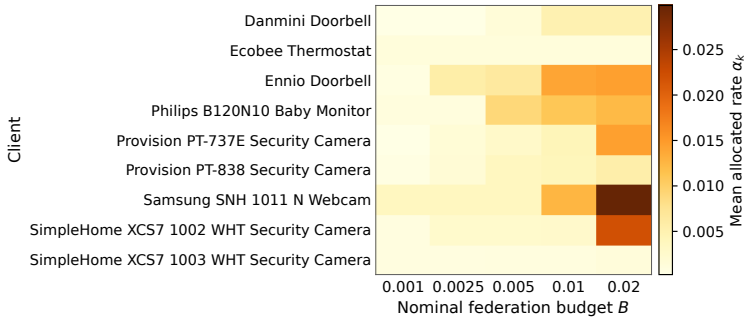


Fig. 2 Mean nominal rates allocated by unrestricted Macro to each N-BaIoT device, averaged across ten original detector-seed runs. The heatmap does not include the 500 frontier-bootstrap resamples.

combination, the reported change frequency is the proportion of resamples whose assigned rate differs from the modal assignment over the 500 resamples. Averaging these frequencies across the ten seeds and nine clients gives 77.8%, 79.8%, 83.9%, 83.6% and 84.9% for ascending nominal budgets. Each value averages 90 client–seed frequencies; it is not a percentage of seeds whose rate changed. The result indicates considerable sensitivity of the optimization decision to the available frontier evidence, without representing repeated operational deployments or changes relative to the single original allocation.

At $B = 0.020$, the Samsung webcam and SimpleHome XCS7 1002 security camera receive the largest mean allocations, whereas Danmini Doorbell receives a smaller allowance.

5.2 Tail objective and worst-client behavior

Tail is optimized for the mean of the three lowest client utilities, yet the observed minimum-client recall is zero for every FABRID-family policy and budget in Table 1. Danmini Doorbell explains this failure of the minimum endpoint: no attacks are detected under any evaluated unrestricted-policy operating point. Its final-calibration and held-out benign FPRs are also zero throughout the 100 combinations of unrestricted policy, seed and budget. Danmini's benign final-calibration sample contains 4,955 records, corresponding to a strict-threshold zero-alert transition at $2/(4{,}955 + 1) = 0.000404$.

Table 2 Prespecified paired seed-level contrasts ($n = 10$): 50,000 paired-bootstrap CIs and Holm-adjusted exact paired sign-flip p values over the five-budget family.

	Macro − Equal-FPR: macro recall			Tail − Equal-FPR: minimum-client recall		
B	Mean Δ	95% CI	Holm p	Mean Δ	95% CI	Holm p
0.0010	0.0000	[0.0000, 0.0000]	1.0000	0.0000	[0.0000, 0.0000]	1.0000
0.0025	−0.0222	[−0.0556, 0.0111]	0.3594	0.0000	[0.0000, 0.0000]	1.0000
0.0050	−0.3800	[−0.4133, −0.3466]	0.0098	0.0000	[0.0000, 0.0000]	1.0000
0.0100	−0.4555	[−0.5000, −0.4111]	0.0098	−0.9998	[−0.9998, −0.9998]	0.0098
0.0200	−0.2556	[−0.2889, −0.2222]	0.0098	−0.9999	[−0.9999, −0.9999]	0.0098

The assigned rates span both sides of that transition, but no evaluated unrestricted threshold recovers positive attack recall for this device.

Ecobee behaves differently. At $B = 0.020$, Macro assigns a mean nominal rate of 0.001164 to Ecobee. Its held-out benign FPR is 0.006672 and its macro attack recall is 0.6000, even though its final-calibration benign FPR is zero.

Calibration sample size affects threshold resolution but does not alone explain the observed difference in test alerts. Since the released partitions are source ordered rather than known independent draws, the experiment cannot quantify how much of Ecobee's discrepancy is due to finite calibration resolution and how much reflects differences between the sampled score distributions.

5.3 Primary contrasts and ranking ceiling

Macro and Equal-FPR are identical at $B = 0.001$; their -0.0222 difference at $B = 0.0025$ is not statistically distinguishable from zero, and Macro recall is lower at larger budgets. Both Tail and Equal-FPR have zero mean minimum-client recall through $B = 0.005$. At $B = 0.010$ and 0.020, Equal-FPR reaches 0.9998 and 0.9999 minimum-client recall while Tail remains at zero (Table 2); Equal-FPR's realized FPR is also $1.68B$ and $1.88B$ at those budgets.

Threshold reallocation cannot change the ordering of the stored anomaly scores, so AUROC and AUPRC are identical across policies within a detector seed. These metrics are computed after pooling the nine devices' held-out benign scores with attack-test scores from their eligible subtypes. The implementation calculates AUROC from score ranks and AUPRC by trapezoidal integration of the precision–recall curve. Averaged over ten seeds, the pooled scores yield AUROC 0.9995 and AUPRC 0.9999. This pooled calculation weights devices and attack subtypes by the number of their retained test rows rather than equally. In particular, the evaluation pool contains 1,041,078 attack rows and 111,188 benign rows, so attacks account for 90.35% of the pool. That prevalence raises the baseline level of precision–recall performance. Excellent pooled scores therefore do not establish equally good score separation for every device or acceptable recall at the selected local thresholds. In the same evaluation, minimum-client recall remains zero for all FABRID-family policy–budget combinations.

The confidence intervals and adjusted p-values in Table 2 do not adjust the recall comparison to a common measured FPR.

5.4 Secondary protocol stress test

The secondary test evaluates unrestricted Macro with ordinary order-statistic calibration on CIC IoT-DIAD. Table 3 summarizes ten separately trained scorers applied to 19 equally weighted physical-device clients, using reconstruction error as the anomaly score.

Table 3 CIC IoT-DIAD: unrestricted Macro with ordinary local calibration; ten-seed means over 19 equally weighted clients using reconstruction-error scores.

B	Macro recall	Minimum-client recall	Realized FPR
0.0010	0.3614	0.0000	0.0098
0.0025	0.3970	0.0000	0.0118
0.0050	0.4318	0.0007	0.0160
0.0100	0.4602	0.0089	0.0213
0.0200	0.5016	0.0251	0.0311

The realized federation FPR exceeds the nominal constraint throughout this secondary test. The overshoot ratios at ascending budgets are 9.80, 4.72, 3.20, 2.13 and 1.56, even as the absolute measured FPR rises from 0.0098 to 0.0311. A decreasing ratio is therefore not evidence that false alarms become less frequent. Unlike the primary experiment, this branch does not include Equal-FPR under the same scorer and clients. It confirms that a nominal-to-measured FPR gap also appears in the secondary configuration, but does not show whether Macro offers better recall at a comparable false-alarm rate.

6 Discussion

The evaluated results do not demonstrate that optimized reallocation is consistently better than the equal-rate baseline at a given measured FPR.

At $B = 0.001$, conservative calibration reduces the measured federation FPR compared with the ordinary rule, but macro attack recall also decreases. Neither calibration choice recovers attack detections for Danmini Doorbell under the tested Macro and Tail variants. All these allocations remain feasible under the nominal constraint, which is why feasibility alone is insufficient as an operational performance claim.

The attack labels used for the utility estimates impose a deployment requirement: FABRID-IDS is an attack-informed allocation procedure applied after training, not an unsupervised online mechanism that discovers suitable client rates without attack examples.

The measured allocation-solver cost is comparatively small: across budgets, Macro runs from 95.7 to 358.8 ms and Tail from 310.3 to 3883.1 ms. None of the recorded optimizations required solver fallback. These figures time the mathematical allocation step only. They exclude the cost of generating anomaly scores, communication, transmitting calibration information and processing alerts. They should therefore be used

to characterize solver execution rather than interpreted as latency measurements for a deployed federated IDS.

Unresolved deployment questions include behavior under adversarial clients, formal privacy guarantees, the total communication and alert-handling cost, and whether improvements to the frozen scorer could address the attacks missed by the weakest devices.

The $B = 0.020$ comparison illustrates why an unqualified recall ranking would be misleading. Equal-FPR reaches macro recall 1.0000 with a measured federation FPR of 1.88B; Macro reaches recall 0.7444 with an FPR of 1.40B (Table 1). The two policies spend different amounts of the actual false-alert allowance, although they were evaluated against the same nominal budget. The result describes those two calibrated operating points and does not determine which policy would be preferable if both were required to operate at an identical measured FPR.

Statistical resolution is limited by the ten paired seeds. With $2^{10} = 1{,}024$ sign assignments, the smallest two-sided exact sign-flip p-value is $2/1{,}024 = 0.001953125$, attained when all ten paired differences are nonzero and have the same sign. With five budget-level tests in a family, the smallest Holm-adjusted value appearing in Table 2 is 0.0098. At $B = 0.001$, the paired policy outcomes are identical, giving $p = 1$ and a zero-width interval. The intervals for minimum-client recall at $B = 0.010$ and 0.020 are also almost degenerate because Tail has zero recall across seeds and Equal-FPR varies only after the fourth decimal place. Holm adjustment is performed within each endpoint family; it does not jointly correct the two families, and the 95% bootstrap intervals remain pointwise.

The 95% statement associated with Eq. (3) applies to one client's population FPR under the required i.i.d. benign sampling assumptions. It is not a simultaneous 95% guarantee for the nine-device federation. Even if the assumptions were valid for each device, applying a union bound to nine separate 5% failure probabilities yields only $1-9(0.05) = 0.55$ as a lower bound on the probability that all nine nominal client limits hold together. The experiments do not establish the required sampling assumptions or an alternative federation-wide confidence construction, so no joint FPR guarantee is claimed for the released N-BaIoT split.

An operational false-alarm ceiling would require thresholds calibrated against benign traffic representative of the intended monitoring period, followed by checks of achieved FPR after deployment. Testing an adaptive monitoring and recalibration mechanism under a credible deployment sequence remains outside the scope of the present source-order evaluation.

Differences in recall must be interpreted alongside realized FPR, because the trade-off depends on the achieved operating point. Tail's lower-group mean adds an independent consideration; a policy can improve two weak clients while leaving the single worst client undetected.

7 Conclusion

FABRID-IDS evaluates how a federation-level nominal false-positive allowance can be distributed across heterogeneous IoT clients after detector training. Local calibration

converts the selected allowances into thresholds, and the resulting measured rates do not always remain within the original federation budget. On N-BaIoT at $B = 0.001$, unrestricted Macro and Tail both reach a measured federation FPR of 0.00170. Conservative calibration reduces this value to 0.00036, but at a cost to attack recall. Equal-FPR records higher macro recall at several nominal budgets while also operating at higher measured FPRs. None of the tested FABRID-family policies restores positive minimum-client recall across the nine-device evaluation. CIC IoT-DIAD shows a similar nominal-to-measured FPR discrepancy under a different configuration, although its experiment lacks a matched Equal-FPR comparator. The results support treating nominal allocation, finite-sample calibration and measured operating performance as separate decisions. Testing whether this framework can maintain a false-alarm ceiling in deployment will require representative calibration streams, monitoring and appropriate distributional assumptions. Implementation and reproducibility artifacts are available at https://doi.org/10.5281/zenodo.22113172. The authors declare no competing interests.

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